

Politecnico di Torino

Collegio di Elettronica, Telecomunicazioni e Fisica

Report for the course
Integrated Systems Architecture

Master degree in Electrical Engineering

Group: 29

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Signature 4: _____

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CHAPTER 1

Lab 2: digital arithmetic

1.1 FPU model

1.1.1 Simulation

The FPU has been stimulated with the following numbers, corresponding to the hexadecimal configuration shown in Table 1.1.

n	a		b		r	
1	7	0x40e0	25	0x41c8	175	0x432f
2	25	0x41c8	128	0x4300	3200	0x4548
3	128	0x4300	-7	0xc0e0	-896	0xc460
4	-7	0xc0e0	-25	0xc1c8	175	0x432f
5	-25	0xc1c8	-128	0xc300	3200	0x4548
6	-128	0xc300	-7	0xc0e0	896	0x4460
7	172	0x432c	-245	0xc375	-42240	0xc725
8	172	0x432c	-45	0xc234	-7744	0xc5f2
9	-118	0xc2ec	54.5	0x425a	-6432	0xc5c9
10	16.5	0x4184	252	0x437c	4160	0x4582
11	135	0x4307	-78.5	0xc29d	-10624	0xc626

Table 1.1: Set of values used for FPU testing

Figure 1.1 shows a snapshot of the simulation. As it can be observed by stimulating the FPU with the numbers shown in Table 1.1, the results are the expected ones.

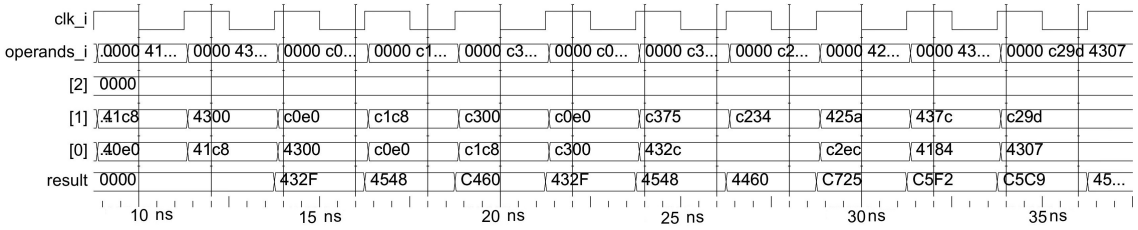


Figure 1.1: Behavioural simulation

1.1.2 Synthesis

In the following figures, there are snapshots of the netlist simulation:

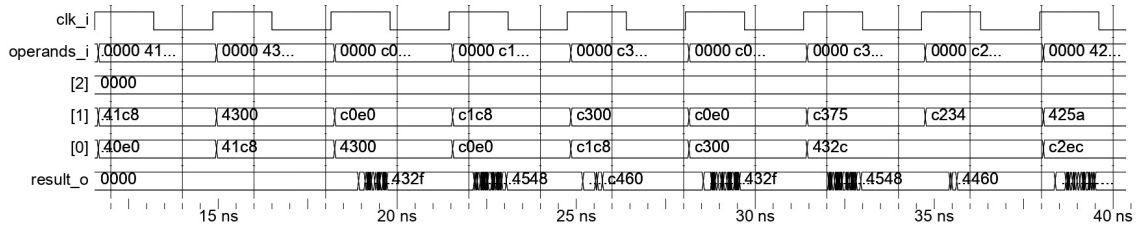


Figure 1.2: Snapshot of netlist simulation

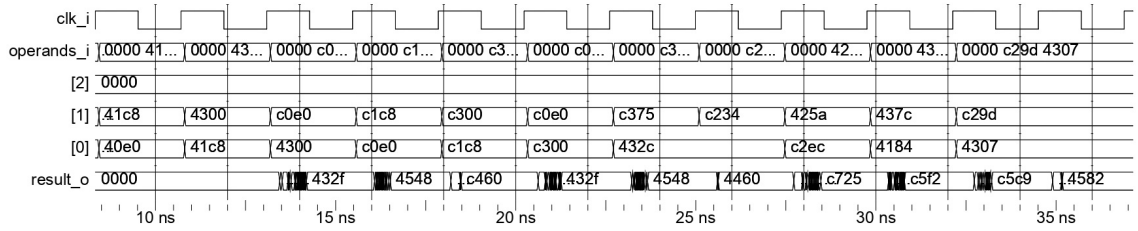


Figure 1.3: Snapshot of netlist simulation with retiming

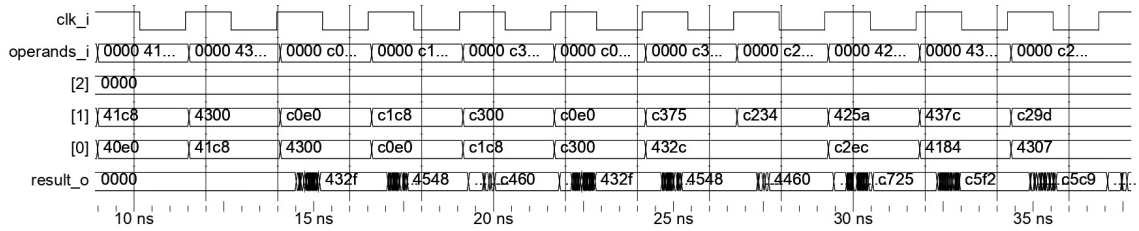


Figure 1.4: Snapshot of netlist simulation with compile ultra

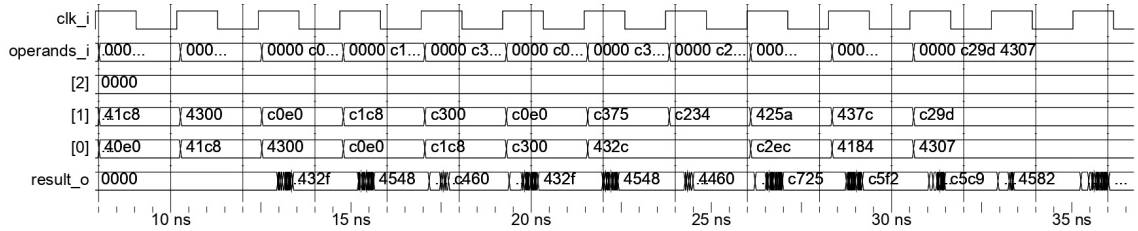


Figure 1.5: Snapshot of netlist simulation with CSA

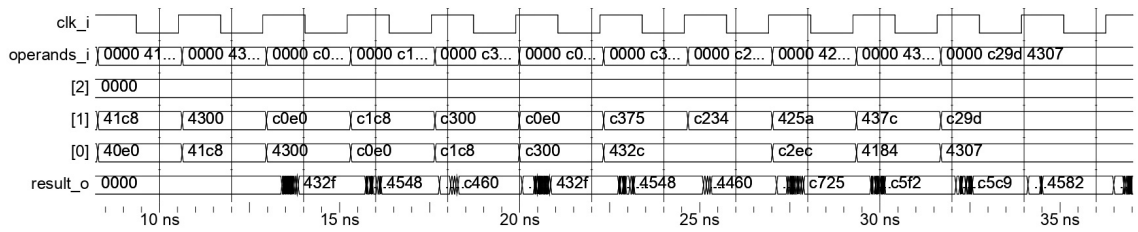


Figure 1.6: Snapshot of netlist simulation with PPRACH

Table 1.2 shows the results of the experiments required in the assignment.
All area and timing reports produced during the synthesis process are included in the appendix.

n	Synthesis strategy	Period [ns]	Frequency [MHz]	Area [μm^2]
1	Base	T_s=3.30	f_M=303	A=3897
2	Retiming	T_s=2.38	f_M=420	A=3580
3	Compile ultra	T_s=2.55	f_M=392	A=3363
4	CSA	T_s=2.26	f_M=442	A=3519
5	PPRACH	T_s=2.35	f_M=425	A=3432

Table 1.2: Main results of all different synthesis

1.1.3 Explanations, comparisons and comments

The results in table 1.1 show the correct working of the given FPU. Moreover, since the precision of Brain Floating Point 16 format is given by the formula $Precision = 2^{-7} \cdot 2^x$(where "x" is the representation of the exponent bits), also the quantization error is verified: for example, values 172 and -245 produce an output of -42140, but since the precision in that range is $2^{-7} \cdot 2^{15} = 256 = \pm 127$, the resulting value becomes -42240, with an error of 100. The same is true for the values: -6431, 4158, -10597.5, which result respectively in -6432, 4160, -10624.

The results of the testbench are compared with a file generated by the C++ executable $fp16_numm$ which creates expected results.
From the table 1.2 it can be seen that, without specifing the architecture used for the mantissa multiplier, the result of the three synthesis with the highest maximum frequency is the 'Retiming' version.

The improvement of maximum frequency between the 'Base' version and the 'Compile ultra' should be guaranteed by the Design Compiler (from the man page of compile_ultra command) and is thus not commented.

The improvement between the 'Compile ultra' version and the 'Retiming' is given by the different syntehsis techinques of the two: the 'Compile ultra' synthesis relies on the 'compile_ultra' command issued to the Design Compiler and uses more advanced algorithms to meet the imposed timing constraints on the design, while the 'Retiming' synthesis uses, besides the 'compile' command, the command 'optimize_registers' which tries to meet the timing requirements by moving the sequential cells in the design. This in turn breaks the critical path of the circuit and ends up having higher maximum frequency than the 'Compile Ultra' synthesis.

When specifying the architecture of the multiplier, both CSA (Carry Save Adder) and PPARCH (Parallel Prefix) return similar maximum frequencies, with the parallel prefix one being 17MHz higher. This result would be compliant with theory since the CSA implementation has an higher delay than the PPARCH one, albeit small, but without knowing the actual implementation of the $DW02_{mult}$ module the answer relies on speculation.

1.2 R4-MBE multiplier

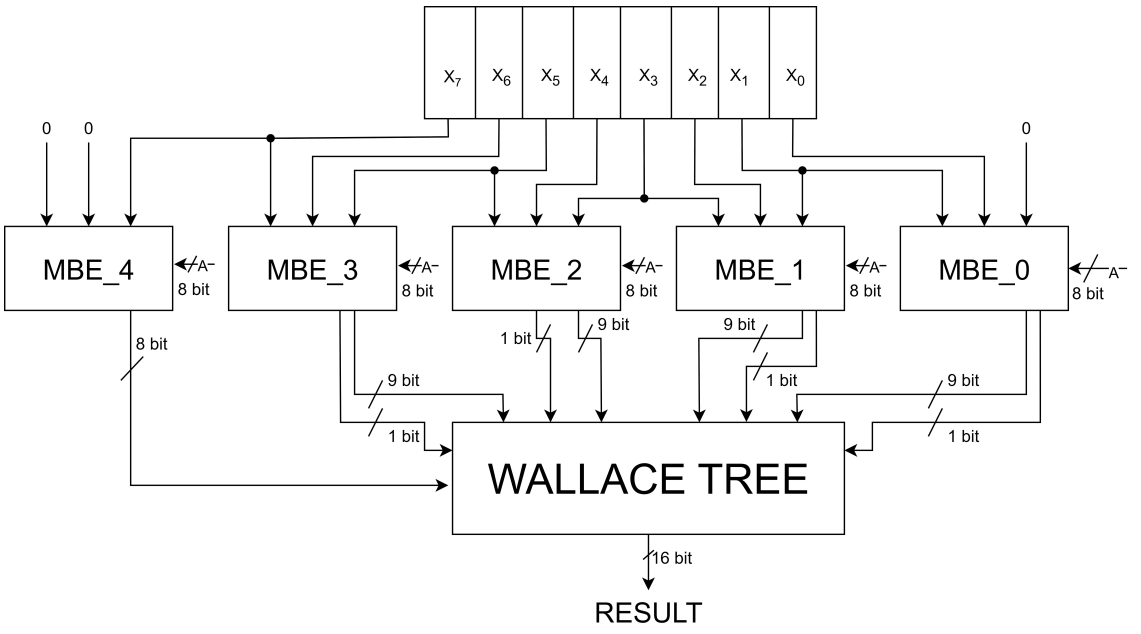


Figure 1.7: R4-MBE architecture

The circuit implementing the R4-MBE with the wallace tree like structure to perform the sum of partial products is shown in Figure 1.7.
The "X" vector is split into 8 bits to generate the three input bits in each modified booth encoder to perform the maximum frequency and produce all partial products in parallel. For this reason five MBE are implemented in the schematic.

The table(1.3) below explain how the X_i bits are connected to each modify booth encoder. Notice that the MBE_4 is connected in a different way. Infact, it receives the input "$00X_7$" and the only possible output is 0 or A, in compliance with the table in figure 1.9.

It is not necessary to add an extra output bit to represent $2A$ and we use only 8 bits. Moreover $-A$ is not a possible output and the bit of sign is not required.

MBE_{ID}	X_{j-1}	X_j	X_{j+1}
MBE0	X_1	X_0	0
MBE 1	X_3	X_2	X_1
MBE 2	X_3	X_4	X_3
MBE 3	X_7	X_6	X_5
MBE 4	0	0	X_7

Table 1.3: MBE input configuration

The image reported in Figure 1.8 illustrates the architecture used to implement the modify booth encoder:
it is implemented using 2 multiplexers and a booth encoder to generate the correct outputs.
Moreover, the first mux is used to select the absolute value, while the second mux selects the 1-complement if needed, since in this architecture the 2-complement is performed in two steps. The output pin RES is the generated partial product and the bit S is the sign of the result. To achieve the correct result the second step of 2-complement conversion (+1) is expected to be applied during the later addition.

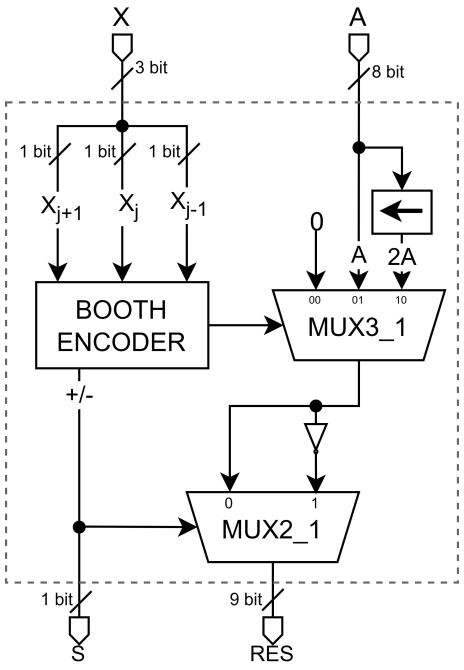
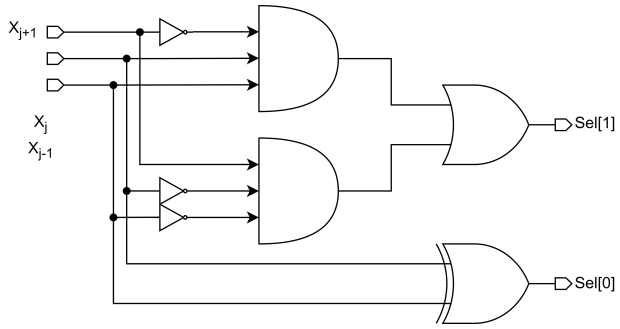


Figure 1.8: Multiplier booth encoding implementation

The Booth encoder logic was implemented by using the corresponding truth table, with the aim of minimizing both the number of logic gates and the resulting latency.
The resulting optimized combinational network is shown in Figure 1.9.



Input			Output		
X_{j+1}	X_j	X_{j-1}	sel[1]	sel[0]	result
0	0	0	0	0	0
0	0	1	0	1	A
0	1	0	0	1	A
0	1	1	1	0	2A
1	0	0	1	0	-2A
1	0	1	0	1	-A
1	1	0	0	1	-A
1	1	1	0	0	0

Figure 1.9: Booth encoder logic implementation

The partial products generated by each MBE block must be accumulated in order to obtain the final multiplication result. This accumulation is performed through a Wallace tree, which takes as input the five initial rows of partial products, as illustrated in Figure 1.10. Where the dots represent the operands and the "S" signals are the sign of each MBE.

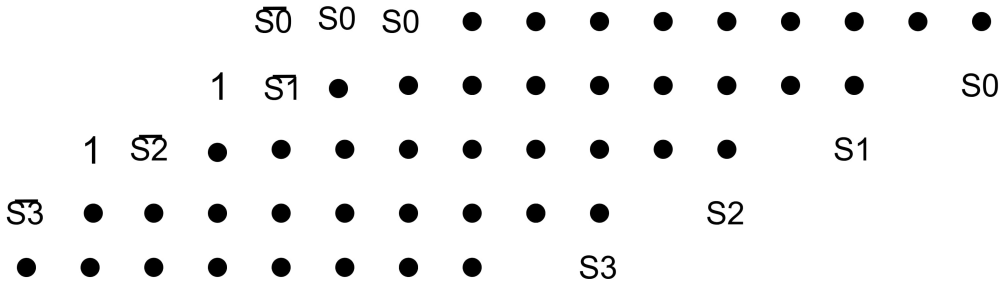


Figure 1.10: Starting point for the Wallace tree

The Wallace tree is composed of four different layers, shown in Figure 1.11, each of which progressively compresses the partial products generated by the modify booth encoder. This hierarchical compression using half-adders and full-adders minimizes the height of the partial-product matrix at each stage. At the end the last layer is composed by a ripple carry adder that produce the final result of multiplication.

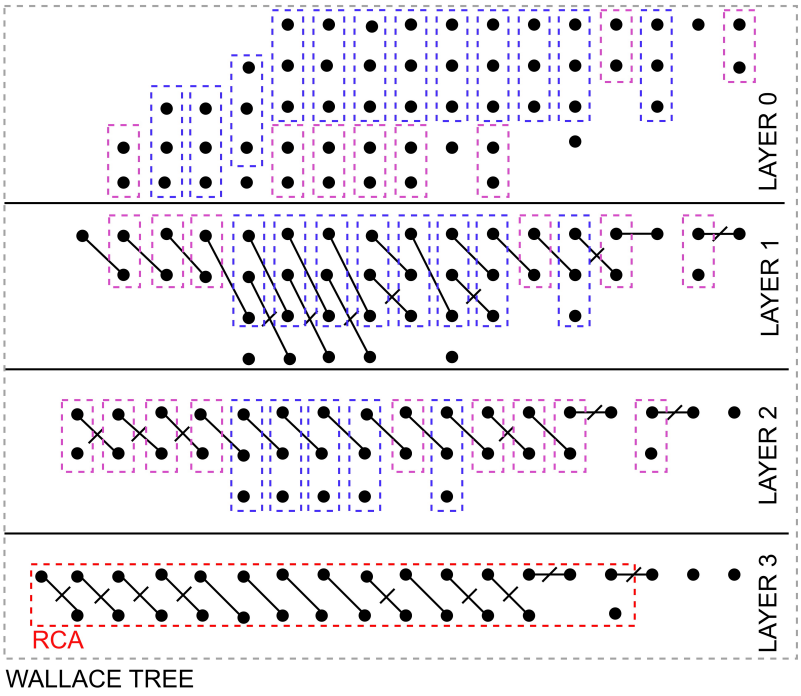


Figure 1.11: Implementation of Wallace tree

1.2.1 Simulation

Stand-alone multiplier Before integrating the custom R4-MBE multiplier into the full FPU datapath, its behaviour was first verified as a stand-alone module. Figure 1.12 shows a snapshot of this preliminary simulation, where the multiplier is stimulated independently to validate the Booth encoding, partial-product generation, and Wallace tree.

The developed testbench provides the following cases to the multiplier:

- multiplication by 1
- multiplication by 0
- multiplication 0 x 0
- multiplication 255 x 255 (all bits to 1)
- 100 random values for the operands

All tests passed.

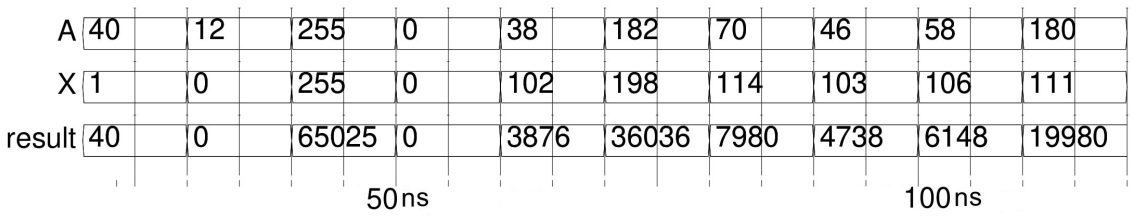


Figure 1.12: Snapshot of edge cases and some random values simulation of stand-alone multiplier

Whole FPU Figure 1.13 shows a snapshot of a simulation for the designed multiplier included in the whole FPU.
After validating the multiplier as an independent module, it was integrated into the complete FPU architecture.
The generated results, reported match the expected BF16 outputs.

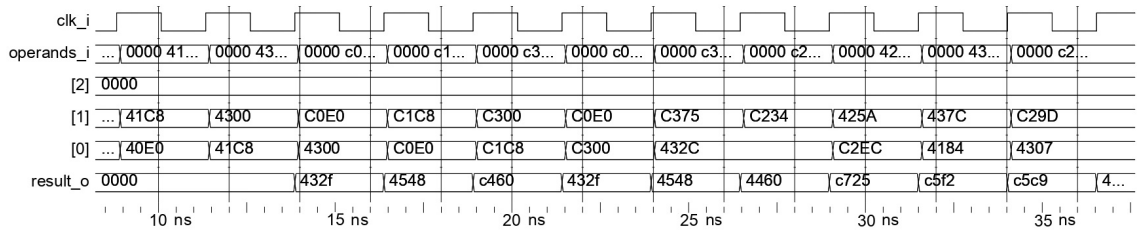


Figure 1.13: Simulation of complete FPU

1.2.2 Synthesis

The results of this synthesis process are reported in Table 1.4.
After synthesis we provide to verify the netlist by simulation, the result is shows in figure 1.14.

Synthesis strategy	Period [ns]	Frequency [MHz]	Area [μm^2]
R4MBE	T_s=2.52	f_M=397	A=3409

Table 1.4: Main results of synthesis with R4MBE

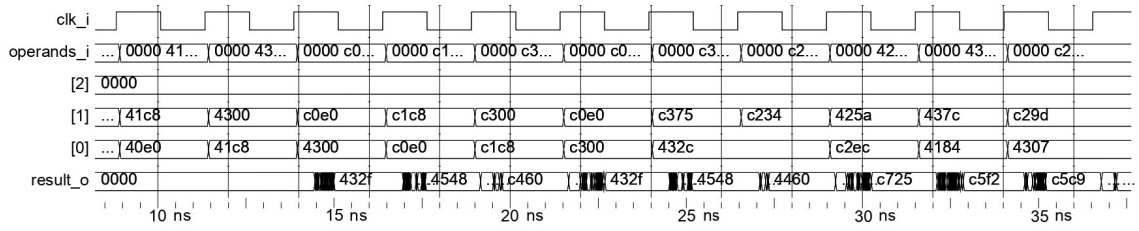


Figure 1.14: Simulation of complete FPU

1.2.3 Explanations, comparisons and comments

As it can be observed from the synthesis results reported in Table 1.4(2.52 ns), the custom R4-MBE multiplier achieves performance figures that are comparable with the synthesis of the compile ultra case in table 1.2(2.55 ns). Overall with the two implementations, behavioral and structural generate the same results in term of frequency and area, this implies that using the behavioral approach $y = a * b$ provides good results with significantly lower effort on the designer side.

The module has been tested with the same values and approach used in section 1.1.3.

1.3 Appendix

1.3.1 Base synthesis report timing

```
Information: Updating design information... (UID-85)

****************************************
Report : timing
        -path full
        -delay max
        -max_paths 1
Design : fpnew_top
Version: S-2021.06-SP4
Date   : Mon Nov 17 15:38:04 2025
****************************************

Operating Conditions: typical   Library: NangateOpenCellLibrary
Wire Load Model Mode: top

  Startpoint: gen_operation_groups[0].i_opgroup_block/gen_parallel_slices
      [4].active_format.i_fmt_slice/gen_num_lanes[0].active_lane.
      lane_instance.i_fma/mid_pipe_sum_q_reg[1][11]
              (rising edge-triggered flip-flop clocked by CLK)
  Endpoint: status_o[UF]
            (output port clocked by CLK)
  Path Group: CLK
  Path Type: max

  Des/Clust/Port     Wire Load Model       Library
  ------------------------------------------------
  fpnew_top          5K_hvratio_1_1        NangateOpenCellLibrary

  Point                                                    Incr       Path
  --
      ------------------------------------------------------------------------

  clock CLK (rise edge)                                    0.00       0.00
  clock network delay (ideal)                              0.00       0.00
  gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
      active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
      i_fma/mid_pipe_sum_q_reg[1][11]/CK (DFFR_X1)
                                                           0.00       0.00 r
  gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
      active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
      i_fma/mid_pipe_sum_q_reg[1][11]/Q (DFFR_X1)
                                                           0.09       0.09 f
  gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
      active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
      i_fma/i_lzc/in_i[11] (lzc_00000013_1)
                                                           0.00       0.09 f
  gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
      active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
      i_fma/i_lzc/U27/ZN (OR3_X2)
                                                           0.09       0.17 f
  gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
      active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
      i_fma/i_lzc/U25/ZN (OR2_X1)
                                                           0.05       0.23 f
  gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
      active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
      i_fma/i_lzc/U35/ZN (NOR2_X1)
                                                           0.04       0.26 r
  gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
      active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
      i_fma/i_lzc/U36/ZN (NAND2_X1)
```

```
                                                          0.03       0.30 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/i_lzc/U29/ZN (AND2_X1)
                                                          0.04       0.33 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/i_lzc/U45/ZN (AND2_X1)
                                                          0.03       0.37 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/i_lzc/U85/ZN (AOI22_X1)
                                                          0.05       0.42 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/i_lzc/U54/ZN (NAND3_X1)
                                                          0.04       0.46 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/i_lzc/cnt_o[1] (lzc_00000013_1)
                                                          0.00       0.46 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U97/ZN (OR2_X1)
                                                          0.06       0.52 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U133/ZN (AND2_X1)
                                                          0.04       0.56 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U131/ZN (NAND2_X1)
                                                          0.03       0.59 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U165/ZN (NAND2_X1)
                                                          0.03       0.61 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U141/ZN (NAND2_X1)
                                                          0.03       0.64 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U167/ZN (NAND2_X1)
                                                          0.03       0.67 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U412/ZN (OAI21_X1)
                                                          0.03       0.71 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U152/ZN (NOR2_X1)
                                                          0.03       0.73 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U155/ZN (XNOR2_X1)
                                                          0.06       0.79 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U416/ZN (OAI211_X1)
                                                          0.04       0.83 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U158/ZN (NOR2_X1)
                                                          0.02       0.85 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
```

```
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U427/ZN (AOI22_X1)
                                                          0.07       0.92 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U162/ZN (OR2_X1)
                                                          0.05       0.96 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U150/ZN (NAND2_X1)
                                                          0.05       1.01 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U260/ZN (OAI222_X1)
                                                          0.06       1.07 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U743/ZN (INV_X1)
                                                          0.04       1.11 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U109/ZN (AND2_X2)
                                                          0.06       1.17 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U783/ZN (NAND2_X1)
                                                          0.05       1.21 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U792/ZN (INV_X1)
                                                          0.02       1.24 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U104/Z (MUX2_X1)
                                                          0.07       1.31 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U793/ZN (OAI222_X1)
                                                          0.06       1.36 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U794/ZN (INV_X1)
                                                          0.03       1.39 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U799/ZN (AOI222_X1)
                                                          0.07       1.46 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U3/ZN (AND2_X2)
                                                          0.06       1.51 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U67/ZN (NAND3_X1)
                                                          0.05       1.56 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U37/Z (BUF_X1)
                                                          0.05       1.61 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U852/ZN (OAI221_X1)
                                                          0.07       1.67 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U853/ZN (OAI221_X1)
```

```
                                                          0.05       1.72 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U860/ZN (NOR4_X1)
                                                          0.06       1.78 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U861/ZN (OAI221_X1)
                                                          0.07       1.85 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U71/ZN (AND2_X1)
                                                          0.05       1.91 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U932/ZN (OAI211_X1)
                                                          0.04       1.95 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/i_fpnew_rounding/abs_value_i[0] (fpnew_rounding_0000000f)
                                                          0.00       1.95 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/i_fpnew_rounding/U50/ZN (NOR2_X1)
                                                          0.03       1.97 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/i_fpnew_rounding/U59/ZN (NOR2_X1)
                                                          0.03       2.01 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/i_fpnew_rounding/U22/Z (MUX2_X1)
                                                          0.05       2.05 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/i_fpnew_rounding/U20/ZN (OAI21_X1)
                                                          0.03       2.08 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/i_fpnew_rounding/U60/ZN (OAI221_X1)
                                                          0.04       2.12 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/i_fpnew_rounding/U16/ZN (AND2_X1)
                                                          0.05       2.18 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/i_fpnew_rounding/U3/ZN (AND3_X1)
                                                          0.06       2.23 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/i_fpnew_rounding/U13/ZN (NAND2_X1)
                                                          0.03       2.26 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/i_fpnew_rounding/U12/ZN (XNOR2_X1)
                                                          0.06       2.32 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/i_fpnew_rounding/abs_rounded_o[12] (fpnew_rounding_0000000f)
                                                          0.00       2.32 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U80/ZN (AND2_X1)
                                                          0.05       2.37 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
```

```
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U123/ZN (NAND3_X1)
                                                          0.03       2.40 f
  gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U1014/ZN (OAI21_X1)
                                                          0.05       2.44 r
  gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U1015/ZN (NAND2_X1)
                                                          0.03       2.47 f
  gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U1016/ZN (NAND2_X1)
                                                          0.03       2.51 r
  gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U128/ZN (AND2_X1)
                                                          0.04       2.54 r
  gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/status_o[UF] (
    fpnew_fma_4_00000002_3__logic_Z_1yB___logic_Z_1yB_)
                                                          0.00       2.54 r
  gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/U38/ZN (AND2_X1)
                                                          0.04       2.58 r
  gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/status_o[UF] (
    fpnew_opgroup_fmt_slice_0_4_00000010_0_00000002_3__logic_Z_1yB_)
                                                          0.00       2.58 r
  gen_operation_groups[0].i_opgroup_block/i_arbiter/data_i[4][status][UF] (
    rr_arb_tree_00000005_1_024242)
                                                          0.00       2.58 r
  gen_operation_groups[0].i_opgroup_block/i_arbiter/U144/ZN (AOI22_X1)
                                                          0.03       2.61 f
  gen_operation_groups[0].i_opgroup_block/i_arbiter/U145/ZN (NAND2_X1)
                                                          0.03       2.64 r
  gen_operation_groups[0].i_opgroup_block/i_arbiter/data_o[status][UF] (
    rr_arb_tree_00000005_1_024242)
                                                          0.00       2.64 r
  gen_operation_groups[0].i_opgroup_block/status_o[UF] (
    fpnew_opgroup_block_0_00000010_0_01_4_155_3__logic_Z_1yB__395949)
                                                          0.00       2.64 r
  i_arbiter/data_i[0][status][UF] (rr_arb_tree_00000004_1_891242)
                                                          0.00       2.64 r
  i_arbiter/U103/ZN (AOI22_X1)                            0.03       2.67 f
  i_arbiter/U104/ZN (NAND2_X1)                            0.04       2.71 r
  i_arbiter/data_o[status][UF] (rr_arb_tree_00000004_1_891242)
                                                          0.00       2.71 r
  status_o[UF] (out)                                      0.02       2.73 r
  data arrival time                                                  2.73

  clock CLK (rise edge)                                   3.30       3.30
  clock network delay (ideal)                             0.00       3.30
  clock uncertainty                                      -0.07       3.23
  output external delay                                  -0.50       2.73
  data required time                                                 2.73
  ---------------------------------------------------------------------------
  data required time                                                 2.73
  data arrival time                                                 -2.73
  ---------------------------------------------------------------------------
  slack (MET)                                                        0.00
1
```

1.3.2 Base synthesis report area

```
****************************************
Report : area
Design : fpnew_top
Version: S-2021.06-SP4
Date   : Mon Nov 17 15:38:04 2025
****************************************

Library(s) Used:

    NangateOpenCellLibrary (File: /eda/dk/nangate45/synopsys/
       NangateOpenCellLibrary_typical_ecsm.db)

Number of ports:                         1986
Number of nets:                          4665
Number of cells:                         2894
Number of combinational cells:           2669
Number of sequential cells:               160
Number of macros/black boxes:               0
Number of buf/inv:                        608
Number of references:                      15

Combinational area:               3043.306003
Buf/Inv area:                      363.621999
Noncombinational area:             853.860027
Macro/Black Box area:                0.000000
Net Interconnect area:      undefined  (Wire load has zero net area)

Total cell area:                  3897.166031
Total area:                 undefined
1
```

1.3.3 Retiming synthesis report timing

```
Information: Updating design information... (UID-85)

****************************************
Report : timing
        -path full
        -delay max
        -max_paths 1
Design : fpnew_top
Version: S-2021.06-SP4
Date   : Mon Nov 17 15:45:51 2025
****************************************

Operating Conditions: typical   Library: NangateOpenCellLibrary
Wire Load Model Mode: top

  Startpoint: gen_operation_groups[0].i_opgroup_block/gen_parallel_slices
      [4].active_format.i_fmt_slice/gen_num_lanes[0].active_lane.
      lane_instance.i_fma/CLK_r_REG126_S2
              (rising edge-triggered flip-flop clocked by CLK)
  Endpoint: status_o[UF]
            (output port clocked by CLK)
  Path Group: CLK
  Path Type: max

  Des/Clust/Port     Wire Load Model       Library
  ------------------------------------------------
  fpnew_top          5K_hvratio_1_1        NangateOpenCellLibrary

  Point                                                    Incr       Path
  --
      ---------------------------------------------------------------------------

  clock CLK (rise edge)                                    0.00       0.00
  clock network delay (ideal)                              0.00       0.00
  gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
      active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
      i_fma/CLK_r_REG126_S2/CK (DFFS_X1)
                                                           0.00       0.00 r
  gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
      active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
      i_fma/CLK_r_REG126_S2/Q (DFFS_X1)
                                                           0.08       0.08 f
  gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
      active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
      i_fma/U197/ZN (XNOR2_X1)
                                                           0.08       0.16 r
  gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
      active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
      i_fma/U318/ZN (OAI33_X1)
                                                           0.05       0.22 f
  gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
      active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
      i_fma/U73/ZN (OR2_X1)
                                                           0.07       0.29 f
  gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
      active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
      i_fma/U327/ZN (OAI222_X4)
                                                           0.12       0.41 r
  gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
      active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
      i_fma/U861/ZN (INV_X1)
                                                           0.06       0.46 f
  gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
      active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
```

```
    i_fma/U20/ZN (AND2_X1)
                                                          0.07       0.53 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U87/ZN (AND2_X1)
                                                          0.05       0.59 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U62/ZN (NOR3_X1)
                                                          0.06       0.65 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U193/ZN (OAI221_X1)
                                                          0.06       0.70 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U991/ZN (INV_X1)
                                                          0.04       0.74 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U81/ZN (NAND3_X1)
                                                          0.05       0.79 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U80/ZN (NAND2_X1)
                                                          0.06       0.84 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U994/ZN (OAI211_X1)
                                                          0.05       0.89 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U326/ZN (AOI22_X1)
                                                          0.05       0.94 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U330/ZN (OAI211_X1)
                                                          0.05       0.99 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U279/ZN (OAI22_X1)
                                                          0.05       1.04 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U137/ZN (INV_X1)
                                                          0.04       1.08 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U1023/ZN (OAI211_X1)
                                                          0.05       1.12 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/i_fpnew_rounding/abs_value_i[0] (fpnew_rounding_0000000f)
                                                          0.00       1.12 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/i_fpnew_rounding/U9/Z (CLKBUF_X1)
                                                          0.06       1.18 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/i_fpnew_rounding/U43/ZN (AND2_X1)
                                                          0.04       1.23 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/i_fpnew_rounding/U44/ZN (AND2_X1)
                                                          0.04       1.27 r
```

```
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/i_fpnew_rounding/U84/ZN (NAND4_X1)
                                                          0.04       1.31 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/i_fpnew_rounding/U17/Z (CLKBUF_X1)
                                                          0.05       1.36 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/i_fpnew_rounding/U85/ZN (INV_X1)
                                                          0.03       1.39 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/i_fpnew_rounding/U35/ZN (NAND2_X1)
                                                          0.03       1.42 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/i_fpnew_rounding/U54/ZN (XNOR2_X1)
                                                          0.06       1.48 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/i_fpnew_rounding/abs_rounded_o[12] (fpnew_rounding_0000000f)
                                                          0.00       1.48 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U173/ZN (NAND3_X1)
                                                          0.04       1.52 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U171/ZN (NOR2_X1)
                                                          0.04       1.56 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U181/ZN (NAND2_X1)
                                                          0.03       1.59 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U166/ZN (NAND2_X1)
                                                          0.03       1.62 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U215/ZN (AND3_X1)
                                                          0.05       1.67 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/status_o[UF] (
    fpnew_fma_4_00000002_3__logic_Z_1yB___logic_Z_1yB_)
                                                          0.00       1.67 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/U5/ZN (AND2_X1)
                                                          0.04       1.70 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/status_o[UF] (
    fpnew_opgroup_fmt_slice_0_4_00000010_0_00000002_3__logic_Z_1yB_)
                                                          0.00       1.70 r
gen_operation_groups[0].i_opgroup_block/i_arbiter/data_i[4][status][UF] (
    rr_arb_tree_00000005_1_024242)
                                                          0.00       1.70 r
gen_operation_groups[0].i_opgroup_block/i_arbiter/U24/ZN (AND2_X1)
                                                          0.04       1.74 r
gen_operation_groups[0].i_opgroup_block/i_arbiter/data_o[status][UF] (
    rr_arb_tree_00000005_1_024242)
                                                          0.00       1.74 r
gen_operation_groups[0].i_opgroup_block/status_o[UF] (
    fpnew_opgroup_block_0_00000010_0_01_4_155_3__logic_Z_1yB__395949)
```

```
                                                          0.00       1.74 r
  i_arbiter/data_i[0][status][UF] (rr_arb_tree_00000004_1_891242)
                                                          0.00       1.74 r
  i_arbiter/U19/ZN (AND2_X1)                              0.04       1.78 r
  i_arbiter/data_o[status][UF] (rr_arb_tree_00000004_1_891242)
                                                          0.00       1.78 r
  status_o[UF] (out)                                      0.02       1.81 r
  data arrival time                                                  1.81

  clock CLK (rise edge)                                   2.38       2.38
  clock network delay (ideal)                             0.00       2.38
  clock uncertainty                                      -0.07       2.31
  output external delay                                  -0.50       1.81
  data required time                                                 1.81
  --
      ------------------------------------------------------------------------

  data required time                                                 1.81
  data arrival time                                                 -1.81
  --
      ------------------------------------------------------------------------

  slack (MET)                                                        0.00


1
```

1.3.4 Retiming synthesis report area

```
****************************************
Report : area
Design : fpnew_top
Version: S-2021.06-SP4
Date   : Mon Nov 17 15:45:52 2025
****************************************

Library(s) Used:

    NangateOpenCellLibrary (File: /eda/dk/nangate45/synopsys/
       NangateOpenCellLibrary_typical_ecsm.db)

Number of ports:                         1933
Number of nets:                          3486
Number of cells:                         2533
Number of combinational cells:           2041
Number of sequential cells:               247
Number of macros/black boxes:               0
Number of buf/inv:                        514
Number of references:                      14

Combinational area:               2266.851997
Buf/Inv area:                      287.546001
Noncombinational area:            1314.040042
Macro/Black Box area:                0.000000
Net Interconnect area:      undefined  (Wire load has zero net area)

Total cell area:                  3580.892040
Total area:                 undefined
1
```

1.3.5 Ultra mode synthesis report timing

```
Information: Updating design information... (UID-85)

****************************************
Report : timing
        -path full
        -delay max
        -max_paths 1
Design : fpnew_top
Version: S-2021.06-SP4
Date   : Mon Nov 17 15:54:38 2025
****************************************

Operating Conditions: typical   Library: NangateOpenCellLibrary
Wire Load Model Mode: top

  Startpoint: gen_operation_groups[0].i_opgroup_block/gen_parallel_slices
      [4].active_format.i_fmt_slice/gen_num_lanes[0].active_lane.
      lane_instance.i_fma/mid_pipe_sum_q_reg[1][17]
              (rising edge-triggered flip-flop clocked by CLK)
  Endpoint: status_o[UF]
            (output port clocked by CLK)
  Path Group: CLK
  Path Type: max

  Des/Clust/Port     Wire Load Model       Library
  ------------------------------------------------
  fpnew_top          5K_hvratio_1_1        NangateOpenCellLibrary

  Point                                                    Incr       Path
  --
      ---------------------------------------------------------------------------

  clock CLK (rise edge)                                    0.00       0.00
  clock network delay (ideal)                              0.00       0.00
  gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
      active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
      i_fma/mid_pipe_sum_q_reg[1][17]/CK (DFFR_X1)
                                                           0.00       0.00 r
  gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
      active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
      i_fma/mid_pipe_sum_q_reg[1][17]/QN (DFFR_X1)
                                                           0.09       0.09 r
  U1452/ZN (NAND2_X1)                                      0.03       0.12 f
  U1160/ZN (NOR2_X1)                                       0.05       0.17 r
  U1639/ZN (AND3_X2)                                       0.07       0.24 r
  U1430/ZN (NAND2_X1)                                      0.04       0.28 f
  U1631/ZN (NAND4_X1)                                      0.04       0.32 r
  U1630/ZN (NAND3_X1)                                      0.04       0.35 f
  U1572/ZN (OR2_X1)                                        0.07       0.42 f
  U1599/ZN (OAI21_X1)                                      0.04       0.46 r
  U1588/ZN (NAND2_X1)                                      0.03       0.49 f
  U1615/ZN (NAND2_X1)                                      0.04       0.53 r
  U1616/ZN (INV_X1)                                        0.03       0.56 f
  U1472/ZN (OR3_X2)                                        0.07       0.63 f
  U1583/ZN (NAND2_X1)                                      0.04       0.68 r
  U1677/ZN (INV_X1)                                        0.03       0.71 f
  U1578/ZN (AND2_X4)                                       0.07       0.78 f
  U1863/ZN (AOI22_X1)                                      0.07       0.85 r
  U983/ZN (AND2_X1)                                        0.05       0.90 r
  U1866/ZN (NAND2_X1)                                      0.03       0.93 f
  U1873/ZN (NAND4_X1)                                      0.03       0.97 r
  U1874/ZN (OAI21_X1)                                      0.03       1.00 f
  U1881/ZN (NAND2_X1)                                      0.04       1.04 r
  U1924/ZN (AND3_X2)                                       0.06       1.09 r
```

```
  U1376/ZN (AND2_X1)                                      0.07       1.16 r
  U1619/ZN (AND2_X1)                                      0.05       1.22 r
  U1608/ZN (OAI21_X1)                                     0.02       1.24 f
  U1110/ZN (AND2_X1)                                      0.04       1.29 f
  U1378/ZN (OR2_X1)                                       0.06       1.35 f
  U1666/ZN (NOR2_X1)                                      0.05       1.40 r
  U1683/ZN (AOI21_X2)                                     0.05       1.45 f
  U1722/ZN (NAND4_X1)                                     0.05       1.50 r
  U1552/ZN (OAI211_X1)                                    0.04       1.54 f
  U1500/ZN (OAI211_X1)                                    0.05       1.59 r
  U1502/ZN (AND2_X2)                                      0.07       1.66 r
  U1523/ZN (NAND4_X1)                                     0.06       1.72 f
  U1554/ZN (INV_X1)                                       0.03       1.75 r
  U1501/ZN (OR2_X1)                                       0.03       1.78 r
  U1519/ZN (AND3_X1)                                      0.05       1.83 r
  U2879/ZN (AND4_X1)                                      0.06       1.89 r
  U1665/ZN (AND4_X2)                                      0.07       1.96 r
  status_o[UF] (out)                                      0.02       1.98 r
  data arrival time                                                  1.98

  clock CLK (rise edge)                                   2.55       2.55
  clock network delay (ideal)                             0.00       2.55
  clock uncertainty                                      -0.07       2.48
  output external delay                                  -0.50       1.98
  data required time                                                 1.98
  --
      ------------------------------------------------------------------------

  data required time                                                 1.98
  data arrival time                                                 -1.98
  --
      ------------------------------------------------------------------------

  slack (MET)                                                        0.00


1
```

1.3.6 Ultra mode synthesis report area

```
****************************************
Report : area
Design : fpnew_top
Version: S-2021.06-SP4
Date   : Mon Nov 17 15:54:38 2025
****************************************

Library(s) Used:

    NangateOpenCellLibrary (File: /eda/dk/nangate45/synopsys/
       NangateOpenCellLibrary_typical_ecsm.db)

Number of ports:                           96
Number of nets:                          2871
Number of cells:                         2633
Number of combinational cells:           2500
Number of sequential cells:               132
Number of macros/black boxes:               0
Number of buf/inv:                        399
Number of references:                      39

Combinational area:               2660.797997
Buf/Inv area:                      223.440001
Noncombinational area:             702.240023
Macro/Black Box area:                0.000000
Net Interconnect area:      undefined  (Wire load has zero net area)

Total cell area:                  3363.038020
Total area:                 undefined
1
```

1.3.7 CSA synthesis report timing

```
Information: Updating design information... (UID-85)

****************************************
Report : timing
        -path full
        -delay max
        -max_paths 1
Design : fpnew_top
Version: S-2021.06-SP4
Date   : Mon Nov 17 16:01:15 2025
****************************************

Operating Conditions: typical   Library: NangateOpenCellLibrary
Wire Load Model Mode: top

  Startpoint: CLK_r_REG168_S1
              (rising edge-triggered flip-flop clocked by CLK)
  Endpoint: CLK_r_REG24_S2
            (rising edge-triggered flip-flop clocked by CLK)
  Path Group: CLK
  Path Type: max

  Des/Clust/Port     Wire Load Model       Library
  ------------------------------------------------
  fpnew_top          5K_hvratio_1_1        NangateOpenCellLibrary

  Point                                                    Incr       Path
  --
      ------------------------------------------------------------------------

  clock CLK (rise edge)                                    0.00       0.00
  clock network delay (ideal)                              0.00       0.00
  CLK_r_REG168_S1/CK (DFFS_X1)                             0.00       0.00 r
  CLK_r_REG168_S1/Q (DFFS_X1)                              0.09       0.09 r
  gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
      active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
      i_fma/sub_293/B[9] (fpnew_top_DW01_sub_6)
                                                           0.00       0.09 r
  gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
      active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
      i_fma/sub_293/U133/Z (XOR2_X1)
                                                           0.07       0.16 r
  gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
      active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
      i_fma/sub_293/DIFF[9] (fpnew_top_DW01_sub_6)
                                                           0.00       0.16 r
  U944/Z (CLKBUF_X1)                                       0.05       0.22 r
  U1402/ZN (OAI21_X1)                                      0.03       0.25 f
  U1403/ZN (INV_X1)                                        0.05       0.30 r
  U1419/ZN (NAND2_X1)                                      0.05       0.35 f
  U1431/ZN (INV_X1)                                        0.05       0.40 r
  U737/ZN (NAND2_X1)                                       0.04       0.44 f
  U1443/ZN (OAI22_X1)                                      0.06       0.49 r
  U1457/Z (MUX2_X1)                                        0.05       0.55 r
  U1458/ZN (INV_X1)                                        0.02       0.57 f
  U1484/Z (MUX2_X1)                                        0.07       0.64 f
  U1485/ZN (INV_X1)                                        0.04       0.68 r
  U1508/ZN (AOI22_X1)                                      0.04       0.72 f
  U1509/Z (XOR2_X1)                                        0.08       0.80 f
  add_1_root_gen_operation_groups[0].i_opgroup_block/gen_parallel_slices
      [4].active_format.i_fmt_slice/gen_num_lanes[0].active_lane.
      lane_instance.i_fma/add_368_2/B[9] (fpnew_top_DW01_add_9)
                                                           0.00       0.80 f
  add_1_root_gen_operation_groups[0].i_opgroup_block/gen_parallel_slices
```

```
    [4].active_format.i_fmt_slice/gen_num_lanes[0].active_lane.
    lane_instance.i_fma/add_368_2/U271/ZN (NOR2_X1)
                                                        0.07       0.87 r
  add_1_root_gen_operation_groups[0].i_opgroup_block/gen_parallel_slices
    [4].active_format.i_fmt_slice/gen_num_lanes[0].active_lane.
    lane_instance.i_fma/add_368_2/U450/ZN (OAI21_X1)
                                                        0.05       0.91 f
  add_1_root_gen_operation_groups[0].i_opgroup_block/gen_parallel_slices
    [4].active_format.i_fmt_slice/gen_num_lanes[0].active_lane.
    lane_instance.i_fma/add_368_2/U288/ZN (AOI21_X1)
                                                        0.07       0.99 r
  add_1_root_gen_operation_groups[0].i_opgroup_block/gen_parallel_slices
    [4].active_format.i_fmt_slice/gen_num_lanes[0].active_lane.
    lane_instance.i_fma/add_368_2/U358/ZN (OAI21_X1)
                                                        0.04       1.02 f
  add_1_root_gen_operation_groups[0].i_opgroup_block/gen_parallel_slices
    [4].active_format.i_fmt_slice/gen_num_lanes[0].active_lane.
    lane_instance.i_fma/add_368_2/U482/ZN (AOI21_X1)
                                                        0.05       1.08 r
  add_1_root_gen_operation_groups[0].i_opgroup_block/gen_parallel_slices
    [4].active_format.i_fmt_slice/gen_num_lanes[0].active_lane.
    lane_instance.i_fma/add_368_2/U285/Z (CLKBUF_X2)
                                                        0.07       1.15 r
  add_1_root_gen_operation_groups[0].i_opgroup_block/gen_parallel_slices
    [4].active_format.i_fmt_slice/gen_num_lanes[0].active_lane.
    lane_instance.i_fma/add_368_2/U316/ZN (OAI21_X1)
                                                        0.05       1.20 f
  add_1_root_gen_operation_groups[0].i_opgroup_block/gen_parallel_slices
    [4].active_format.i_fmt_slice/gen_num_lanes[0].active_lane.
    lane_instance.i_fma/add_368_2/SUM[28] (fpnew_top_DW01_add_9)
                                                        0.00       1.20 f
  U1539/ZN (INV_X1)                                     0.03       1.23 r
  U1540/ZN (NAND3_X1)                                   0.04       1.27 f
  U1082/ZN (INV_X2)                                     0.09       1.36 r
  U1057/Z (BUF_X1)                                      0.11       1.46 r
  U1573/ZN (AOI22_X1)                                   0.04       1.51 f
  U1574/ZN (OAI21_X1)                                   0.04       1.55 r
  U2080/ZN (INV_X1)                                     0.03       1.58 f
  U874/Z (BUF_X1)                                       0.04       1.62 f
  U946/ZN (AND3_X1)                                     0.05       1.67 f
  U750/ZN (AND2_X1)                                     0.04       1.71 f
  U887/ZN (OR2_X1)                                      0.06       1.77 f
  U1161/ZN (OAI221_X1)                                  0.07       1.84 r
  U1602/ZN (INV_X1)                                     0.03       1.87 f
  U1191/ZN (OAI21_X1)                                   0.04       1.92 r
  U1190/ZN (OAI21_X1)                                   0.03       1.95 f
  U1606/ZN (OAI21_X1)                                   0.04       1.99 r
  U1189/ZN (AOI21_X1)                                   0.04       2.02 f
  U909/ZN (OR2_X1)                                      0.06       2.09 f
  U1034/ZN (XNOR2_X1)                                   0.05       2.14 f
  CLK_r_REG24_S2/D (DFFS_X1)                            0.01       2.15 f
  data arrival time                                                2.15

  clock CLK (rise edge)                                 2.26       2.26
  clock network delay (ideal)                           0.00       2.26
  clock uncertainty                                    -0.07       2.19
  CLK_r_REG24_S2/CK (DFFS_X1)                           0.00       2.19 r
  library setup time                                   -0.04       2.15
  data required time                                               2.15
  ------------------------------------------------------------------------
  data required time                                               2.15
  data arrival time                                               -2.15
  ------------------------------------------------------------------------
  slack (MET)                                                      0.00
1
```

1.3.8 CSA synthesis report area

```
****************************************
Report : area
Design : fpnew_top
Version: S-2021.06-SP4
Date   : Mon Nov 17 16:01:15 2025
****************************************

Library(s) Used:

    NangateOpenCellLibrary (File: /eda/dk/nangate45/synopsys/
       NangateOpenCellLibrary_typical_ecsm.db)

Number of ports:                          462
Number of nets:                          2719
Number of cells:                         2244
Number of combinational cells:           1971
Number of sequential cells:               253
Number of macros/black boxes:               0
Number of buf/inv:                        489
Number of references:                      47

Combinational area:               2173.485995
Buf/Inv area:                      273.980001
Noncombinational area:            1345.960043
Macro/Black Box area:                0.000000
Net Interconnect area:      undefined  (Wire load has zero net area)

Total cell area:                  3519.446038
Total area:                 undefined
1
```

1.3.9 CSA synthesis report resources

```
****************************************
Report : resources
Design : fpnew_top
Version: S-2021.06-SP4
Date   : Mon Nov 17 16:01:15 2025
****************************************

Resource Sharing Report for design fpnew_top in file ../src/fpnew_top.sv

================================================================================

|          |              |            | Contained     |
                        |
| Resource | Module       | Parameters | Resources     | Contained
   Operations |
================================================================================

| r426     | DW01_cmp2    | width=3    |               |
   gen_operation_groups[0].i_opgroup_block/i_arbiter/gt_208_G4 |
|               |                        |
|          |              |            |               |
   gen_operation_groups[0].i_opgroup_block/i_arbiter/lte_209_G4 |
| r427     | DW01_cmp2    | width=3    |               |
   gen_operation_groups[1].i_opgroup_block/i_arbiter/gt_208_G4 |
|               |                        |
|          |              |            |               |
   gen_operation_groups[1].i_opgroup_block/i_arbiter/lte_209_G4 |
| r428     | DW01_cmp2    | width=3    |               |
   gen_operation_groups[2].i_opgroup_block/i_arbiter/gt_208_G4 |
|               |                        |
|          |              |            |               |
   gen_operation_groups[2].i_opgroup_block/i_arbiter/lte_209_G4 |
| r429     | DW01_cmp2    | width=3    |               |
   gen_operation_groups[3].i_opgroup_block/i_arbiter/gt_208_G4 |
|               |                        |
|          |              |            |               |
   gen_operation_groups[3].i_opgroup_block/i_arbiter/lte_209_G4 |
| r430     | DW01_cmp2    | width=2    |               | i_arbiter/
   gt_208_G4  |
|          |              |            |               | i_arbiter/
   lte_209_G4 |
| r460     | DW01_add     | width=10   |               |
   gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
   active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
   i_fma/add_285 |
| r468     | DW01_sub     | width=10   |               |
   gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
   active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
   i_fma/sub_293 |
| r470     | DW_cmp       | width=10   |               |
   gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
   active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
   i_fma/gt_295 |
| r472     | DW_cmp       | width=10   |               |
   gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
   active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
   i_fma/lte_302 |
| r474     | DW_cmp       | width=10   |               |
   gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
   active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
   i_fma/lte_305 |
| r476     | DW01_sub     | width=5    |               |
   gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
```

```
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/sub_306 |
| r478     | DW02_mult    | A_width=8  |               |
    gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/mult_325 |
|
|          |              | B_width=8  |               |
                        |
| r480     | DW_rightsh   | A_width=36 |               |
    gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/srl_349 |
|
|          |              | SH_width=5 |               |
                       |
| r482     | DW01_add     | width=29   |               |
    add_1_root_gen_operation_groups[0].i_opgroup_block/gen_parallel_slices
    [4].active_format.i_fmt_slice/gen_num_lanes[0].active_lane.
    lane_instance.i_fma/add_368_2 |
| r484     | DW01_sub     | width=28   |               |
    gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/sub_372 |
| r486     | DW_cmp       | width=10   |               |
    gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/lte_510 |
| r488     | DW_cmp       | width=10   |               |
    gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/lte_510_2 |
| r494     | DW_cmp       | width=12   |               |
    gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/gte_512 |
| r496     | DW01_add     | width=5    |               |
    gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/add_514 |
| r502     | DW01_add     | width=5    |               |
    gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/add_519 |
| r504     | DW_leftsh    | A_width=29 |               |
    gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/sll_530 |
|
|          |              | SH_width=5 |               |
                       |
| r506     | DW_cmp       | width=10   |               |
    gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/gt_547 |
| r508     | DW01_inc     | width=10   |               |
    gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/add_542 |
| r510     | DW01_dec     | width=10   |               |
    gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/sub_549 |
| r512     | DW_cmp       | width=10   |               |
    gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
```

```
    i_fma/gte_576 |
| r514     | DW01_add     | width=15   |               |
    gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/i_fpnew_rounding/add_63 |
| r1242    | DW01_add     | width=10   |               |
    add_1_root_add_1_root_gen_operation_groups[0].i_opgroup_block/
    gen_parallel_slices[4].active_format.i_fmt_slice/gen_num_lanes[0].
    active_lane.lane_instance.i_fma/sub_287 |
| r1244    | DW01_sub     | width=10   |               |
    sub_2_root_add_1_root_gen_operation_groups[0].i_opgroup_block/
    gen_parallel_slices[4].active_format.i_fmt_slice/gen_num_lanes[0].
    active_lane.lane_instance.i_fma/sub_287 |
| r1246    | DW01_add     | width=10   |               |
    add_0_root_add_1_root_gen_operation_groups[0].i_opgroup_block/
    gen_parallel_slices[4].active_format.i_fmt_slice/gen_num_lanes[0].
    active_lane.lane_instance.i_fma/sub_287 |
| r1966    | DW01_sub     | width=11   |               |
    sub_1_root_gen_operation_groups[0].i_opgroup_block/gen_parallel_slices
    [4].active_format.i_fmt_slice/gen_num_lanes[0].active_lane.
    lane_instance.i_fma/add_512 |
| r1968    | DW01_inc     | width=12   |               |
    add_0_root_gen_operation_groups[0].i_opgroup_block/gen_parallel_slices
    [4].active_format.i_fmt_slice/gen_num_lanes[0].active_lane.
    lane_instance.i_fma/add_512 |
| r2688    | DW01_sub     | width=10   |               |
    sub_1_root_gen_operation_groups[0].i_opgroup_block/gen_parallel_slices
    [4].active_format.i_fmt_slice/gen_num_lanes[0].active_lane.
    lane_instance.i_fma/add_515 |
| r2690    | DW01_inc     | width=10   |               |
    add_0_root_gen_operation_groups[0].i_opgroup_block/gen_parallel_slices
    [4].active_format.i_fmt_slice/gen_num_lanes[0].active_lane.
    lane_instance.i_fma/add_515 |
================================================================================


Implementation Report
================================================================================

|                    |                  | Current            | Set
             |
| Cell               | Module           | Implementation     |
    Implementation |
================================================================================

| gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/add_542 |
|                    | DW01_inc         | pparch (area,speed)
                  |
| add_1_root_add_1_root_gen_operation_groups[0].i_opgroup_block/
    gen_parallel_slices[4].active_format.i_fmt_slice/gen_num_lanes[0].
    active_lane.lane_instance.i_fma/sub_287 |
|                    | DW01_add         | pparch (area,speed)
                 |
| gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/sub_372 |
|                    | DW01_sub         | pparch (speed)     |
                |
| add_1_root_gen_operation_groups[0].i_opgroup_block/gen_parallel_slices
    [4].active_format.i_fmt_slice/gen_num_lanes[0].active_lane.
    lane_instance.i_fma/add_368_2 |
|                    | DW01_add         | pparch (speed)     |
                  |
```

```
| gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/sub_293 |
|                     | DW01_sub          | pparch (area,speed)
                     |
| add_0_root_gen_operation_groups[0].i_opgroup_block/gen_parallel_slices
    [4].active_format.i_fmt_slice/gen_num_lanes[0].active_lane.
    lane_instance.i_fma/add_515 |
|                     | DW01_inc          | rpl                |
                    |
| gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/mult_325 |
|                     | DW02_mult         | csa                | csa
                |
==============================================================================


1
```

1.3.10 PPARCH synthesis report timing

```
Information: Updating design information... (UID-85)

****************************************
Report : timing
        -path full
        -delay max
        -max_paths 1
Design : fpnew_top
Version: S-2021.06-SP4
Date   : Mon Nov 17 16:06:44 2025
****************************************

Operating Conditions: typical   Library: NangateOpenCellLibrary
Wire Load Model Mode: top

  Startpoint: CLK_r_REG84_S1
              (rising edge-triggered flip-flop clocked by CLK)
  Endpoint: CLK_r_REG10_S2
            (rising edge-triggered flip-flop clocked by CLK)
  Path Group: CLK
  Path Type: max

  Des/Clust/Port     Wire Load Model       Library
  ------------------------------------------------
  fpnew_top          5K_hvratio_1_1        NangateOpenCellLibrary

  Point                                                    Incr       Path
  --
      ------------------------------------------------------------------------

  clock CLK (rise edge)                                    0.00       0.00
  clock network delay (ideal)                              0.00       0.00
  CLK_r_REG84_S1/CK (DFFR_X1)                              0.00       0.00 r
  CLK_r_REG84_S1/Q (DFFR_X1)                               0.11       0.11 r
  U777/ZN (AND2_X1)                                        0.04       0.15 r
  U968/ZN (NAND2_X1)                                       0.02       0.17 f
  U1346/ZN (NAND2_X1)                                      0.04       0.22 r
  U1347/ZN (OAI21_X1)                                      0.04       0.26 f
  U1348/ZN (INV_X1)                                        0.05       0.31 r
  U1364/ZN (NAND2_X1)                                      0.05       0.36 f
  U1377/ZN (INV_X1)                                        0.04       0.40 r
  U1038/ZN (NAND2_X1)                                      0.05       0.45 f
  U1378/ZN (OAI22_X1)                                      0.06       0.51 r
  U1379/ZN (INV_X1)                                        0.02       0.53 f
  U1381/Z (MUX2_X1)                                        0.07       0.61 f
  U1405/Z (MUX2_X1)                                        0.08       0.68 f
  U1406/ZN (INV_X1)                                        0.04       0.72 r
  U1462/ZN (AOI22_X1)                                      0.04       0.76 f
  U1463/Z (XOR2_X1)                                        0.08       0.85 f
  add_1_root_gen_operation_groups[0].i_opgroup_block/gen_parallel_slices
      [4].active_format.i_fmt_slice/gen_num_lanes[0].active_lane.
      lane_instance.i_fma/add_368_2/B[5] (fpnew_top_DW01_add_8)
                                                           0.00       0.85 f
  add_1_root_gen_operation_groups[0].i_opgroup_block/gen_parallel_slices
      [4].active_format.i_fmt_slice/gen_num_lanes[0].active_lane.
      lane_instance.i_fma/add_368_2/U283/ZN (NOR2_X2)
                                                           0.07       0.92 r
  add_1_root_gen_operation_groups[0].i_opgroup_block/gen_parallel_slices
      [4].active_format.i_fmt_slice/gen_num_lanes[0].active_lane.
      lane_instance.i_fma/add_368_2/U440/ZN (OAI21_X1)
                                                           0.04       0.95 f
  add_1_root_gen_operation_groups[0].i_opgroup_block/gen_parallel_slices
      [4].active_format.i_fmt_slice/gen_num_lanes[0].active_lane.
      lane_instance.i_fma/add_368_2/U361/ZN (AOI21_X1)
```

```
                                                          0.05       1.00 r
add_1_root_gen_operation_groups[0].i_opgroup_block/gen_parallel_slices
    [4].active_format.i_fmt_slice/gen_num_lanes[0].active_lane.
    lane_instance.i_fma/add_368_2/U284/ZN (OAI21_X1)
                                                          0.04       1.04 f
add_1_root_gen_operation_groups[0].i_opgroup_block/gen_parallel_slices
    [4].active_format.i_fmt_slice/gen_num_lanes[0].active_lane.
    lane_instance.i_fma/add_368_2/U268/ZN (INV_X1)
                                                          0.07       1.11 r
add_1_root_gen_operation_groups[0].i_opgroup_block/gen_parallel_slices
    [4].active_format.i_fmt_slice/gen_num_lanes[0].active_lane.
    lane_instance.i_fma/add_368_2/U495/ZN (OAI21_X1)
                                                          0.04       1.15 f
add_1_root_gen_operation_groups[0].i_opgroup_block/gen_parallel_slices
    [4].active_format.i_fmt_slice/gen_num_lanes[0].active_lane.
    lane_instance.i_fma/add_368_2/U395/ZN (XNOR2_X1)
                                                          0.07       1.22 f
add_1_root_gen_operation_groups[0].i_opgroup_block/gen_parallel_slices
    [4].active_format.i_fmt_slice/gen_num_lanes[0].active_lane.
    lane_instance.i_fma/add_368_2/SUM[9] (fpnew_top_DW01_add_8)
                                                          0.00       1.22 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/sub_372/B[9] (fpnew_top_DW01_sub_7)
                                                          0.00       1.22 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/sub_372/U230/ZN (NOR2_X1)
                                                          0.06       1.28 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/sub_372/U138/ZN (NAND2_X1)
                                                          0.04       1.32 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/sub_372/U224/ZN (NOR2_X1)
                                                          0.04       1.36 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/sub_372/U117/ZN (NAND2_X1)
                                                          0.07       1.44 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/sub_372/U181/ZN (NOR2_X1)
                                                          0.06       1.50 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/sub_372/U180/ZN (XNOR2_X1)
                                                          0.03       1.53 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/sub_372/DIFF[18] (fpnew_top_DW01_sub_7)
                                                          0.00       1.53 f
U1496/ZN (AOI22_X1)                                       0.05       1.58 r
U1497/ZN (OAI21_X1)                                       0.04       1.62 f
U2035/ZN (INV_X1)                                         0.04       1.66 r
U1065/ZN (AND2_X1)                                        0.05       1.71 r
U1125/ZN (AND3_X1)                                        0.06       1.76 r
U821/ZN (AND3_X1)                                         0.05       1.82 r
U813/ZN (AND2_X2)                                         0.05       1.87 r
U817/ZN (INV_X1)                                          0.03       1.90 f
U1543/ZN (OAI21_X1)                                       0.06       1.96 r
U1544/ZN (INV_X1)                                         0.04       2.00 f
U950/ZN (OR2_X1)                                          0.07       2.06 f
U866/ZN (AND2_X1)                                         0.04       2.10 f
U1562/ZN (OAI21_X1)                                       0.04       2.14 r
```

```
  U902/ZN (NAND2_X1)                                      0.03       2.17 f
  U1116/ZN (OR2_X1)                                       0.06       2.23 f
  CLK_r_REG10_S2/D (DFFS_X1)                              0.01       2.24 f
  data arrival time                                                  2.24

  clock CLK (rise edge)                                   2.35       2.35
  clock network delay (ideal)                             0.00       2.35
  clock uncertainty                                      -0.07       2.28
  CLK_r_REG10_S2/CK (DFFS_X1)                             0.00       2.28 r
  library setup time                                     -0.04       2.24
  data required time                                                 2.24
  --
    ------------------------------------------------------------------------

  data required time                                                 2.24
  data arrival time                                                 -2.24
  --
    ------------------------------------------------------------------------

  slack (MET)                                                        0.00


1
```

1.3.11 PPARCH synthesis report area

```
****************************************
Report : area
Design : fpnew_top
Version: S-2021.06-SP4
Date   : Mon Nov 17 16:06:44 2025
****************************************

Library(s) Used:

    NangateOpenCellLibrary (File: /eda/dk/nangate45/synopsys/
       NangateOpenCellLibrary_typical_ecsm.db)

Number of ports:                          416
Number of nets:                          2603
Number of cells:                         2146
Number of combinational cells:           1884
Number of sequential cells:               246
Number of macros/black boxes:               0
Number of buf/inv:                        465
Number of references:                      45

Combinational area:               2123.743997
Buf/Inv area:                      262.808001
Noncombinational area:            1308.720042
Macro/Black Box area:                0.000000
Net Interconnect area:      undefined  (Wire load has zero net area)

Total cell area:                  3432.464039
Total area:                 undefined
1
```

1.3.12 PPARCH synthesis report resources

```
****************************************
Report : resources
Design : fpnew_top
Version: S-2021.06-SP4
Date   : Mon Nov 17 16:06:44 2025
****************************************

Resource Sharing Report for design fpnew_top in file ../src/fpnew_top.sv

================================================================================

|          |              |            | Contained     |
                        |
| Resource | Module       | Parameters | Resources     | Contained
   Operations |
================================================================================

| r427     | DW01_cmp2    | width=3    |               |
   gen_operation_groups[0].i_opgroup_block/i_arbiter/gt_208_G4 |
|               |                        |
|          |              |            |               |
   gen_operation_groups[0].i_opgroup_block/i_arbiter/lte_209_G4 |
| r428     | DW01_cmp2    | width=3    |               |
   gen_operation_groups[1].i_opgroup_block/i_arbiter/gt_208_G4 |
|               |                        |
|          |              |            |               |
   gen_operation_groups[1].i_opgroup_block/i_arbiter/lte_209_G4 |
| r429     | DW01_cmp2    | width=3    |               |
   gen_operation_groups[2].i_opgroup_block/i_arbiter/gt_208_G4 |
|               |                        |
|          |              |            |               |
   gen_operation_groups[2].i_opgroup_block/i_arbiter/lte_209_G4 |
| r430     | DW01_cmp2    | width=3    |               |
   gen_operation_groups[3].i_opgroup_block/i_arbiter/gt_208_G4 |
|               |                        |
|          |              |            |               |
   gen_operation_groups[3].i_opgroup_block/i_arbiter/lte_209_G4 |
| r431     | DW01_cmp2    | width=2    |               | i_arbiter/
   gt_208_G4  |
|          |              |            |               | i_arbiter/
   lte_209_G4 |
| r461     | DW01_add     | width=10   |               |
   gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
   active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
   i_fma/add_285 |
| r469     | DW01_sub     | width=10   |               |
   gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
   active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
   i_fma/sub_293 |
| r471     | DW_cmp       | width=10   |               |
   gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
   active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
   i_fma/gt_295 |
| r473     | DW_cmp       | width=10   |               |
   gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
   active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
   i_fma/lte_302 |
| r475     | DW_cmp       | width=10   |               |
   gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
   active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
   i_fma/lte_305 |
| r477     | DW01_sub     | width=5    |               |
   gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
```

```
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/sub_306 |
| r479     | DW02_mult    | A_width=8  |               |
    gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/mult_325 |
|
|          |              | B_width=8  |               |
                        |
| r481     | DW_rightsh   | A_width=36 |               |
    gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/srl_349 |
|
|          |              | SH_width=5 |               |
                       |
| r483     | DW01_add     | width=29   |               |
    add_1_root_gen_operation_groups[0].i_opgroup_block/gen_parallel_slices
    [4].active_format.i_fmt_slice/gen_num_lanes[0].active_lane.
    lane_instance.i_fma/add_368_2 |
| r485     | DW01_sub     | width=28   |               |
    gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/sub_372 |
| r487     | DW_cmp       | width=10   |               |
    gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/lte_510 |
| r489     | DW_cmp       | width=10   |               |
    gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/lte_510_2 |
| r495     | DW_cmp       | width=12   |               |
    gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/gte_512 |
| r497     | DW01_add     | width=5    |               |
    gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/add_514 |
| r503     | DW01_add     | width=5    |               |
    gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/add_519 |
| r505     | DW_leftsh    | A_width=29 |               |
    gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/sll_530 |
|
|          |              | SH_width=5 |               |
                       |
| r507     | DW_cmp       | width=10   |               |
    gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/gt_547 |
| r509     | DW01_inc     | width=10   |               |
    gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/add_542 |
| r511     | DW01_dec     | width=10   |               |
    gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/sub_549 |
| r513     | DW_cmp       | width=10   |               |
    gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
```

```
    i_fma/gte_576 |
| r515     | DW01_add     | width=15   |               |
    gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/i_fpnew_rounding/add_63 |
| r1243    | DW01_add     | width=10   |               |
    add_1_root_add_1_root_gen_operation_groups[0].i_opgroup_block/
    gen_parallel_slices[4].active_format.i_fmt_slice/gen_num_lanes[0].
    active_lane.lane_instance.i_fma/sub_287 |
| r1245    | DW01_sub     | width=10   |               |
    sub_2_root_add_1_root_gen_operation_groups[0].i_opgroup_block/
    gen_parallel_slices[4].active_format.i_fmt_slice/gen_num_lanes[0].
    active_lane.lane_instance.i_fma/sub_287 |
| r1247    | DW01_add     | width=10   |               |
    add_0_root_add_1_root_gen_operation_groups[0].i_opgroup_block/
    gen_parallel_slices[4].active_format.i_fmt_slice/gen_num_lanes[0].
    active_lane.lane_instance.i_fma/sub_287 |
| r1967    | DW01_sub     | width=11   |               |
    sub_1_root_gen_operation_groups[0].i_opgroup_block/gen_parallel_slices
    [4].active_format.i_fmt_slice/gen_num_lanes[0].active_lane.
    lane_instance.i_fma/add_512 |
| r1969    | DW01_inc     | width=12   |               |
    add_0_root_gen_operation_groups[0].i_opgroup_block/gen_parallel_slices
    [4].active_format.i_fmt_slice/gen_num_lanes[0].active_lane.
    lane_instance.i_fma/add_512 |
| r2689    | DW01_sub     | width=10   |               |
    sub_1_root_gen_operation_groups[0].i_opgroup_block/gen_parallel_slices
    [4].active_format.i_fmt_slice/gen_num_lanes[0].active_lane.
    lane_instance.i_fma/add_515 |
| r2691    | DW01_inc     | width=10   |               |
    add_0_root_gen_operation_groups[0].i_opgroup_block/gen_parallel_slices
    [4].active_format.i_fmt_slice/gen_num_lanes[0].active_lane.
    lane_instance.i_fma/add_515 |
================================================================================


Implementation Report
================================================================================

|                    |                  | Current            | Set
             |
| Cell               | Module           | Implementation     |
    Implementation |
================================================================================

| gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/add_542 |
|                    | DW01_inc         | pparch (area,speed)
                  |
| add_1_root_add_1_root_gen_operation_groups[0].i_opgroup_block/
    gen_parallel_slices[4].active_format.i_fmt_slice/gen_num_lanes[0].
    active_lane.lane_instance.i_fma/sub_287 |
|                    | DW01_add         | pparch (area,speed)
                 |
| gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/sub_372 |
|                    | DW01_sub         | pparch (speed)     |
                |
| add_1_root_gen_operation_groups[0].i_opgroup_block/gen_parallel_slices
    [4].active_format.i_fmt_slice/gen_num_lanes[0].active_lane.
    lane_instance.i_fma/add_368_2 |
|                    | DW01_add         | pparch (speed)     |
                  |
```

```
| gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/sub_293 |
|                     | DW01_sub          | pparch (area,speed)
                     |
| add_0_root_gen_operation_groups[0].i_opgroup_block/gen_parallel_slices
    [4].active_format.i_fmt_slice/gen_num_lanes[0].active_lane.
    lane_instance.i_fma/add_515 |
|                     | DW01_inc          | rpl                |
                    |
| gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/mult_325 |
|                     | DW02_mult         | pparch (area)      | pparch
            |
|                     |                   | mult_arch: and     |
                    |
================================================================================


1
```

1.3.13 R4MBE synthesis report timing

```
Information: Updating design information... (UID-85)

****************************************
Report : timing
        -path full
        -delay max
        -max_paths 1
Design : fpnew_top
Version: S-2021.06-SP4
Date   : Sun Nov 16 12:36:15 2025
****************************************

Operating Conditions: typical   Library: NangateOpenCellLibrary
Wire Load Model Mode: top

  Startpoint: gen_operation_groups[0].i_opgroup_block/gen_parallel_slices
      [4].active_format.i_fmt_slice/gen_num_lanes[0].active_lane.
      lane_instance.i_fma/mid_pipe_exp_prod_q_reg[1][2]
              (rising edge-triggered flip-flop clocked by CLK)
  Endpoint: status_o[UF]
            (output port clocked by CLK)
  Path Group: CLK
  Path Type: max

  Des/Clust/Port     Wire Load Model       Library
  ------------------------------------------------
  fpnew_top          5K_hvratio_1_1        NangateOpenCellLibrary

  Point                                                    Incr       Path
  --
      ---------------------------------------------------------------------------

  clock CLK (rise edge)                                    0.00       0.00
  clock network delay (ideal)                              0.00       0.00
  gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
      active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
      i_fma/mid_pipe_exp_prod_q_reg[1][2]/CK (DFFR_X1)
                                                           0.00       0.00 r
  gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
      active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
      i_fma/mid_pipe_exp_prod_q_reg[1][2]/Q (DFFR_X1)
                                                           0.11       0.11 r
  U1290/ZN (AND2_X1)                                       0.05       0.17 r
  U2025/CO (HA_X1)                                         0.06       0.23 r
  U2026/CO (HA_X1)                                         0.06       0.29 r
  U2027/ZN (NOR2_X1)                                       0.03       0.31 f
  U2028/ZN (INV_X1)                                        0.03       0.35 r
  U1169/ZN (AND3_X2)                                       0.05       0.40 r
  U1437/ZN (NAND2_X1)                                      0.03       0.43 f
  U1681/ZN (NAND2_X1)                                      0.03       0.46 r
  U1175/ZN (AND2_X2)                                       0.05       0.51 r
  U1849/ZN (OR2_X1)                                        0.04       0.55 r
  U1373/ZN (NAND2_X1)                                      0.03       0.58 f
  U1375/ZN (INV_X1)                                        0.02       0.60 r
  U1374/ZN (AND3_X2)                                       0.07       0.67 r
  U2048/ZN (NOR2_X1)                                       0.04       0.71 f
  U1196/Z (BUF_X2)                                         0.06       0.76 f
  U2049/ZN (NAND2_X1)                                      0.04       0.80 r
  U1882/ZN (NAND4_X1)                                      0.05       0.85 f
  U2065/ZN (NAND2_X1)                                      0.03       0.89 r
  U2074/ZN (NAND3_X1)                                      0.03       0.92 f
  U2095/ZN (OAI211_X1)                                     0.05       0.97 r
  U1341/ZN (AND3_X2)                                       0.06       1.03 r
  U1249/ZN (AND2_X1)                                       0.08       1.11 r
```

```
  U1918/ZN (NAND2_X1)                                      0.04       1.15 f
  U2204/ZN (NAND2_X1)                                      0.03       1.18 r
  U2205/ZN (NAND2_X1)                                      0.02       1.20 f
  U1960/ZN (NAND2_X1)                                      0.04       1.24 r
  U2206/ZN (NAND4_X1)                                      0.06       1.29 f
  U1305/ZN (OAI21_X1)                                      0.06       1.36 r
  U1298/ZN (NAND2_X2)                                      0.06       1.41 f
  U1809/ZN (INV_X1)                                        0.05       1.47 r
  U2424/ZN (OR2_X1)                                        0.05       1.51 r
  U1456/ZN (AND2_X1)                                       0.04       1.55 r
  U1771/ZN (AND3_X1)                                       0.05       1.60 r
  U1886/ZN (AND3_X2)                                       0.07       1.67 r
  U2428/Z (MUX2_X1)                                        0.07       1.74 r
  U1206/ZN (AND3_X1)                                       0.06       1.80 r
  U1889/ZN (NAND2_X1)                                      0.03       1.83 f
  U1892/ZN (OAI21_X1)                                      0.05       1.88 r
  U1904/ZN (AND2_X1)                                       0.05       1.93 r
  status_o[UF] (out)                                       0.02       1.95 r
  data arrival time                                                   1.95

  clock CLK (rise edge)                                    2.52       2.52
  clock network delay (ideal)                              0.00       2.52
  clock uncertainty                                       -0.07       2.45
  output external delay                                   -0.50       1.95
  data required time                                                  1.95
  --
      ------------------------------------------------------------------------

  data required time                                                  1.95
  data arrival time                                                  -1.95
  --
      ------------------------------------------------------------------------

  slack (MET)                                                         0.00


1
```

1.3.14 R4MBE synthesis report area

```
****************************************
Report : area
Design : fpnew_top
Version: S-2021.06-SP4
Date   : Sun Nov 16 12:36:15 2025
****************************************

Library(s) Used:

    NangateOpenCellLibrary (File: /eda/dk/nangate45/synopsys/
       NangateOpenCellLibrary_typical_ecsm.db)

Number of ports:                           96
Number of nets:                          2929
Number of cells:                         2730
Number of combinational cells:           2597
Number of sequential cells:               132
Number of macros/black boxes:               0
Number of buf/inv:                        444
Number of references:                      43

Combinational area:               2706.549993
Buf/Inv area:                      256.158001
Noncombinational area:             702.240023
Macro/Black Box area:                0.000000
Net Interconnect area:      undefined  (Wire load has zero net area)

Total cell area:                  3408.790015
Total area:                 undefined
1
```